

DHRUV GUPTA

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Summary

Final-year B. Tech CSE student focused on Agentic AI, RAG systems, and LLM-powered automation. Hands-on with LangGraph, LangChain, ChromaDB, Supabase Vector DB, Groq/OpenAI APIs, FastAPI, Node.js, Twilio, and n8n. Interested in building reliable LLM workflows by improving retrieval quality, debugging failure modes, and evaluating outputs using tools like RAGAS.

Education

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar

2023 - Present

B. Tech (Computer Science & Engineering)

- CGPA: 9.46

Delhi Public School Ghaziabad

2019 - 2021

Secondary and Senior Secondary

- Percentage (XII): 93.8

Skills

Programming Languages: Python, JavaScript, C, Java

GenAI & Agentic AI: LLMs, RAG (Retrieval-Augmented Generation), AI Agents, Prompt Engineering, LangChain, LangGraph

AI/ML & Data: Embeddings, Vector Search, NLP, Model Evaluation (RAGAS), Sentence Transformers

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs

Databases & Storage: MongoDB, ChromaDB, Supabase (Vector DB)

Tools & Platforms: OpenAI API, Groq API, Twilio, n8n, ElevenLabs, Git, Streamlit

Concepts: Agentic Workflows, Tool Calling, Memory Systems

Projects

Agentic RAG System for Reliable Physics QA

04/2026

- LangGraph, Groq API, LLaMA 3.3 70B, ChromaDB, RAGAS, Streamlit | [GitHub](#)
- Built a RAG-based physics QA system using ChromaDB for semantic retrieval and Groq API for LLM-based response generation.
- Used LangGraph state management to maintain query context across routing, retrieval, and response-generation steps.
- Used retrieval as a callable tool and added fallback handling for low-context or out-of-scope queries.
- Evaluated the prototype on a small 10-query test set using RAGAS, achieving 0.82 faithfulness and 0.87 answer relevance during local testing.
- Built a Streamlit interface to test queries and inspect retrieval/output behavior.

Multimodal AI Personal Assistant

07/2025

- Built a multimodal AI assistant prototype for text, audio, image, and document-based inputs.
- Implemented a RAG pipeline using OpenAI embeddings and Supabase Vector DB for document-based answering.
- Integrated WhatsApp access through Twilio webhooks during prototype testing.
- Tested contextual response and document-retrieval workflows, noting limitations around ambiguous inputs and incomplete context.

Carbonly – AI Powered Sustainability Analytics Platform

03/2026

- Node.js, Express.js, FastAPI, MongoDB, JWT, bcrypt | [GitHub](#)
- Built a full-stack platform to estimate and track user carbon emissions from lifestyle inputs.
- Developed Node.js/Express APIs for authentication, dashboard workflows, and emission record management.
- Integrated a FastAPI-based ML inference service for carbon emission estimation.
- Tested core API flows and resolved integration issues between backend inputs and the ML inference pipeline.

Achievements

- GATE DA 2026 – AIR 1102